Thus, our original equation changes as follows:

$$f(r) = \sum_{k\in\{0,1\}^\ell} t_k \cdot \widetilde{\mathrm{eq}}(k,r) = \sum_{i\in[\sqrt{m}]}\sum_{j\in[\sqrt{m}]} T_{i,j}\cdot u_i \cdot v_j = \sum_{j\in[\sqrt{m}]}\left(\sum_{i\in[\sqrt{m}]} T_{i,j}\cdot u_i\right)\cdot v_j = \langle w, v\rangle$$

where $w = (w_0, \dots, w_{\sqrt{m}-1})$ and $w_j = \sum_{i\in[\sqrt{m}]} T_{i,j}\cdot u_i$.

Observe that for each $i \in [\sqrt{m}]$, a Pedersen commitment to w_i can be obtained by raising c_i to the exponent u_i. In particular,

$$c_{w_i} = c_i^{u_i} = \left(\prod_{j=0}^{\sqrt{m}-1} g_j^{T_{i,j}}\right)^{u_i} = \prod_{j=0}^{\sqrt{m}-1} g_j^{T_{i,j}\cdot u_i}$$

In the original Hyrax, the prover runs a Bulletproof protocol instance to prove the inner product $\langle w, v\rangle = f(r)$ using a commitment to w_i. In the BabyHyrax, instead, we send the w to the verifier as it is (assuming that the protocol does not need to be zero-knowledge or that it will be used in a zero-knowledge environment).

2.3.2 Sona

So, for an ℓ-variate multilinear polynomial $f\colon \{0,1\}^\ell \to \mathbb{F}$ the commitment and the evaluation proof consists of:

$$C_f = \mathsf{Hash}(c_0, \dots, c_{\sqrt{m}-1}) \in \mathbb{F}$$
$$w = (w_0, \dots, w_{\sqrt{m}-1}) \in \mathbb{F}^{\sqrt{m}}$$

The original Sona leverages the Nova backend [KST22] to prove that the following relation holds for the arbitrary point r and prover answer $v^\star = f(r)$:

$$\mathcal{R} = \left\{\begin{aligned} & C_f = \mathsf{Hash}(c_0, \dots, c_{\sqrt{m}-1}) \\ & \prod_{i\in[\sqrt{m}]} c_i^{u_i} = \prod_{j\in[\sqrt{m}]} g_j^{w_j} \\ & v^\star = \langle w, v\rangle \end{aligned}\right\}$$

Following the definition of Sona commitment protocol, we can describe a bunch of functions that leverage an arbitrary collision-resistant hash function $\mathsf{Hash}\colon \mathbb{G}^{\sqrt{m}} \to \mathbb{F}$, a Nova prover ProveNova and a verifier VerifyNova functions, and generator points $g \in \mathbb{G}^{\sqrt{m}}$:

CommitSona($f\colon \{0,1\}^\ell \to \mathbb{F}$)

1: Put $m = 2^\ell$
2: Evaluate witness vector $t \in \mathbb{F}^m$ from f and restructure it into the matrix T.
3: **for** $i \leftarrow 0$ **to** $\sqrt{m}-1$ **do**
4: $\quad c_i \leftarrow \prod_{j=0}^{\sqrt{m}-1} g_j^{T_{i,j}} \in \mathbb{G}$
5: **end for**
6: $C_f \leftarrow \mathsf{Hash}(c_0, \dots, c_{\sqrt{m}-1})$
7: **return** C_f

OpenSona($C_f \in \mathbb{F}$, $r \in \{0,1\}^\ell$)

1: Put $m = 2^\ell$
2: For each $k \in \{0,1\}^{\ell/2}$ derive u and v according to BabyHyrax:

$$u_k = \prod_{i=0}^{\ell/2-1} (k_i r_i + (1-k_i)(1-r_i))$$